

Henri Cerio-Cain

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EDUCATION

Dalhousie University

Bachelor of Computer Science

Halifax, NS

Expected Sep 2027

- Relevant Coursework: Software Engineering, Probability I/II, Algorithms, Optimization, Linear Programming

WORK EXPERIENCE

Domane

Founding Software Engineer

Jan 2025 – Present

- Built and launched Domane into private TestFlight beta as founding engineer, developing an offline-first React Native/Expo fitness app with TypeScript, Firebase, SQLite, Zustand, and Expo Router
- Implemented production-grade mobile features including workout logging, analytics dashboards, iOS Live Activities/Dynamic Island, and CI-backed Jest/architecture tests

J.D. Irving, Limited

Saint John, NB

Data Scientist Intern

Apr 2026 – Sep 2026

- Building end-to-end analytics pipelines by sourcing, cleaning, and modeling enterprise data with Python, SQL, and Azure to support automation and data-driven decision-making
- Applying exploratory data analysis, feature engineering, and model evaluation to machine learning and statistical use cases in operational settings
- Supporting deployment and adoption of analytics tools through technical documentation, user training, and KPI tracking tied to model and business performance

AI Innovation Student

Jan 2026 – Apr 2026

- Built AI agents in Copilot Studio and Azure integrating enterprise REST APIs to automate cross-system workflows across multiple data sources
- Led feature development in Azure DevOps, collaborating with stakeholders to define requirements and manage user stories, epics, and sprint delivery
- Contributed to a Gurobi mixed-integer optimization model for demand-driven production scheduling, designing constraints and model logic that enabled \$100K+ annual cost savings

SKILLS

- **Languages:** JavaScript, Python, Java, C, SQL, HTML/CSS, Android (Java, Kotlin, XML, MVVM)
- **Machine Learning & Data:** PyTorch, TensorFlow, Hugging Face Transformers, OCR, Model Fine-Tuning
- **Tools & Platforms:** AWS, Azure, Firebase, Git, GitLab, Android Studio, Google Colab, Cloudflare, Uvicorn

PROJECTS

DocAI Organizer

Sep 2025 – Dec 2025

- Built a serverless document organization web app (AWS) that uses Google Gemini 2.5 Flash to classify and organize PDFs into custom folder structures, achieving sub-30-second processing with JWT authentication, bcrypt password hashing, and a cost-optimized architecture at \$1.27/month for 1000 users

Document Routing with an RNN

Sep 2025 – Oct 2025

- Built an OCR-to-text document classification model using Tesseract and a BiLSTM network, achieving ~83% test accuracy across four document types with real-time prediction and confidence scoring for unseen scanned pages

DalTutor

May 2025 – Sep 2025

- Rebuilt an Android tutoring marketplace with a community tab, real-time availability, Google Maps tutor search, PayPal checkout, and CI-tested community moderation tools using atomic multi-path writes and role-based Security Rules for reliable collaboration

HenriAI

Dec 2024 – Aug 2025

- Fine-tuned a 6B-parameter GPT-J model using LoRA and 4-bit QLoRA on a single A100 GPU and deployed a FastAPI inference server with authentication and Cloudflare tunneling at henriai.ca for custom conversational text generation

ADDITIONAL

Languages: French and English (fluent), Cebuano (intermediate)

Certifications: New Brunswick French Oral Proficiency – Advanced